

ARGHA PRATIM SAHA

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LinkedIn | Google Scholar | GitHub

RESEARCH INTERESTS

Human-centered evaluation of AI systems; cybersecurity and privacy; empirical and experimental methods for interactive and agentic systems; behavioral evaluation of large language models. I am interested in how people and AI systems behave under realistic, adversarial, or high-stakes conditions, and in designing rigorous empirical evaluations – controlled experiments, benchmarks, and annotation schemes – to measure that behavior.

EDUCATION

- **BSc. in Computer Science and Engineering** Feb 2020 – July 2025
Shahjalal University of Science and Technology (SUST), Sylhet
CGPA: 3.71/4.00
Thesis: *CyQured: Design, Development, and Empirical Evaluation of a Tabletop Game for Personal Cybersecurity Education*
Supervisors: Dr. Farida Chowdhury, Dr. Md. Sadek Ferdous, Dr. Md. Masum
- **Higher Secondary Certificate (HSC), Science** June 2019
BCIC College, Mirpur, Dhaka | GPA: 5.00/5.00
- **Secondary School Certificate (SSC), Science** May 2017
BCIC School, Mirpur, Dhaka | GPA: 5.00/5.00

PUBLICATIONS

1. **CyQured: Design, Development, and Empirical Evaluation of a Tabletop Game for Personal Cybersecurity Education**
Utsho Das, Argha Pratim Saha*, Md. Sadek Ferdous, Md. Masum, Farida Chowdhury* (*equal contribution)
USENIX Symposium on Usable Privacy and Security (SOUPS), 2026 – Accepted
2. **A Deep Learning Approach to Automate Classification of Arsenic-Affected Skin using EfficientNet-B1**
Argha Pratim Saha, Utsho Das, M. Shahidur Rahman
27th Int. Conference on Computer and Information Technology (ICCIT), Dec 2024
3. **Project PRIMUS at EHRSQL 2024: Text-to-SQL Generation using Large Language Model for EHR Analysis**
Sourav Joy, Rohan Ahmed, Argha Saha, Minhaj Habil, Utsho Das, Partha Bhowmik [\[Paper\]](#)
6th Clinical Natural Language Processing Workshop, NAACL 2024, Mexico City, Mexico

RESEARCH EXPERIENCE

- **Research Assistant (Part-time)** Jan 2026 – Present
BRAC University, Dhaka | Supervisor: Dr. Farida Chowdhury
 - **Usable Security Research:** Investigating how young adults in Bangladesh understand and respond to phishing and smishing attacks; designing low-cognitive-overhead interventions to improve threat-detection confidence.
 - **Study Design:** Owning literature review, research design, participant-facing study materials, and evaluation planning end-to-end.
 - **Academic Mentorship:** Mentoring multiple undergraduate thesis groups through problem formulation, literature evaluation, methodology design, and manuscript writing.
- **Undergraduate Thesis Research** Jun 2024 – Jul 2025
Shahjalal University of Science and Technology (SUST), Sylhet
 - **Intervention Design:** Identified mental-model gaps in consumer cybersecurity and built *CyQured*, a tabletop serious game mapping the STRIDE threat framework to smart-home scenarios for non-expert users.
 - **Controlled Evaluation:** Ran a preliminary calibration study followed by a controlled mixed-methods user study ($n = 50$) using a pre-/post-test design.

- **Statistical Analysis:** Analyzed pre-/post-test data with linear mixed-effects models; measured usability, acceptance, and workload via SUS, TAM, and NASA-TLX. Found statistically significant gains in threat identification alongside high perceived usability.

PROFESSIONAL EXPERIENCE

- **Associate Software Engineer** *Oct 2025 – Present*
ShellBeeHaken Ltd., Dhaka
 - **Full-Stack Engineering (KriyaKarak):** Developed core front-end modules including expert-profile systems and reservation flows; executed end-to-end testing across the stack.
 - **AI Voice Agent (R&D):** Designed a Bengali-English code-switching framework for a real-time conversational agent; integrated a retrieval-augmented generation (RAG) pipeline to reduce context drift.
 - **Real-Time Infrastructure:** Built a WebSocket-based pipeline for low-latency, bi-directional voice streaming between users and the agent.

RESEARCH METHODS

- **Study Design:** Controlled/RCT-style user studies, pre-/post-test designs, questionnaire-based evaluation, mixed-methods research.
- **Analysis:** Descriptive and inferential statistics, linear mixed-effects models, Python/R-based analysis.
- **Evaluation:** SUS, TAM, NASA-TLX, annotation schema design, usability evaluation.
- **Research Practice:** Literature review, research-question formulation, academic writing, participant-facing study preparation.

TECHNICAL SKILLS

- **Programming Languages:** Python, C/C++, Go, JavaScript, TypeScript, Java
- **AI & ML:** Machine Learning, Deep Learning, Large Language Models, Text-to-SQL, Retrieval-Augmented Generation (RAG), Speech-to-Text, Text-to-Speech
- **Web & Systems:** React, Next.js, Node.js, Express.js, FastAPI, WebSockets, REST APIs
- **Data & Tools:** MySQL, MongoDB, Qdrant, R, Git, GitHub, L^AT_EX

SELECTED PROJECTS

- **AutoScriber** [[GitHub](#)] – Automated audio-to-text pipeline for qualitative HCI researchers working with Bengali interviews.
 - Custom audio preprocessing to isolate speech, remove background noise, and trim silences.
 - Integrated local open-source speech models alongside cloud APIs to improve Bengali transcription accuracy. *TypeScript, Node.js*.
- **git-user** [[GitHub](#)] – Cross-platform Go utility for managing and switching between multiple Git identities and encrypted SSH key pairs.
 - Isolated credential pool secured with passphrase-based encryption.
 - Published on npm as [[git-userhub](#)] with **3,000+** downloads in its first month. *Go, Shell, JavaScript*.

AFFILIATIONS & ACADEMIC SERVICE

Member, Human-Centered Computing Society (HCCS) *Jan 2026 – Present*
BRAC University, Dhaka | [HCCS website](#)
Engagement in digital accessibility workshops and cross-institutional HCI initiatives.

PROBLEM SOLVING & COMPETITIVE PROGRAMMING

- Solved 1,000+ algorithmic problems across online judges (LeetCode, Codeforces, LightOJ, HackerRank, HackerEarth, VJudge).
- **LeetCode:** [bob_dylan](#) – contest rating 1900+, 360+ problems solved.
 - **Codeforces:** [fugitive_stranger](#) – max rating 1419 (Specialist).
 - **Contests:** ICPC Bangladesh Preliminary 2021; 5th place, AppifyLab Programming Contest 2025; multiple intra-university contests.